

Mathematical Induction I

def. The standard proof technique in computer science. ^{"hypotheses"}

- Used to check conjectures about the outcomes of processes that occur repeatedly & according to a definite pattern.

Format: For all integers $\{ \text{condition using } n \}$, $P(n)$ is true, where $P(n)$ is the sentence "_____".

- An argument by mathematical induction is one that first proves the condition/equation of K , and then proves $K+1$.

PRINCIPLE-

Let $P(n)$ be a property defined for integers of n , and let a be a fixed integer. Suppose the following statements are true:

- $P(a)$ is true.
- For all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true.

Then for all integers $n \geq a$, $P(n)$ is true.

- The validity of proof by mathematical induction is taken as an axiom, which is why it's called a principle instead of a theorem.
↓ "a statement regarded as self-evidently true"

Example of MI-

In a row of dominoes, if the k th domino falls back, then the $(k+1)$ domino also falls back.

$P(n)$ = "The n th domino falls back."

It's given that for each $k \geq 1$, if $P(k)$ is true the domino falls back, then $P(k+1)$ is also true. It's also given that $P(1)$, first domino falls, is also true.

Therefore, $P(n)$ is true for every integer $n \geq 1$.

METHOD OF PROOF

Step 1/Basis Step. Show the property for $n=a$ is true.

Step 2/Inductive Step. Show that for all integers $k \geq a$, if $n=k$ is true then it is true for $n=k+1$.

• To perform the inductive step, suppose that the property for $n=k$ is true, where k is any integer ~~except~~ with $k \geq a$.

▪ k is an arbitrarily chosen ~~#~~

• Then show the property is also true for $n=k+1$
✓

Example - Sum of the First n Integers

Use mathematical induction to prove that

$$1 + 2 + \dots + n = \frac{n(n+1)}{2} \quad \text{for all integers } n \geq 1.$$

$$P(n) \Rightarrow 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$P(n)$ = The sum of all integers from 1 to n equals $\frac{n(n+1)}{2}$.

Basis Step. prove $n=a$

$$\text{If } n=1, P(1) \text{ is } 1 = \frac{1(1+1)}{2} = \frac{1(2)}{2} = 1.$$

The property is true for $n=1$

Inductive Step. prove $n=k+1$

$$\text{If } k=1, P(k+1) \text{ or } P(1+1) \text{ is } 1+2 = \frac{2(2+1)}{2} = \frac{2(3)}{2} = 3.$$

The example formatted is...

For all integers $n \geq 1$,

$$1+2+\dots+n = \frac{n(n+1)}{2}.$$

Let the property $P(n)$ be the equation $1+2+\dots+n = \frac{n(n+1)}{2}$.

To prove the property for $n=1$, we must show

$1 = \frac{1(1+1)}{2}$. The right side of the equation does simplify to 1, so $P(1)$ is true.

If the property is true for all integers or $n = k \geq 1$, then it is true for $n = k+1$.

Suppose $1+2+\dots+k = \frac{k(k+1)}{2}$, for some integer $k \geq 1$.
(inductive hypothesis) ↗

Then, $1+2+\dots+(k+1) = \frac{(k+1)(k+1+1)}{2}$.

$$\begin{array}{ccc} \text{Let } k=3 : & 1+2+3+(3+1) = & \frac{(3+1)(3+1+1)}{2} \\ & \downarrow & \downarrow \frac{4(5)}{2} = 10 \\ & 10 & \end{array}$$

Therefore, the property $P(n)$ is true for any integer $n \geq 1$.